

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA02

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CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

1. Low Contrast Image

(a) Interpretation: The image's pixel intensities are compressed into a narrow band, so objects and background produce nearly identical gray values. This is typically caused by poor lighting during capture, limited sensor dynamic range, or a naturally low-contrast scene.

(b) Key issues:

- Histogram concentrated in a small intensity range
- Objects and background are visually indistinguishable
- Reduced effectiveness of later processing such as thresholding or segmentation
- Loss of perceptual/structural detail

(c) Applied technique — Histogram Equalization (or linear Contrast Stretching): Redistribute intensities using the cumulative distribution function so the histogram is spread across the full 0–255 range, boosting global contrast.

(d) Justification: Histogram equalization needs no prior knowledge of the image's true intensity range and works robustly on any dull/low-contrast image; contrast stretching is preferred instead when the min/max intensities are known and a simple linear mapping is sufficient.

(e) Summary: Cause → narrow histogram range; Fix → equalization/stretching; Result → intensities spread across the full range, giving clearly visible objects against the background.

2. Uneven Illumination

(a) Interpretation: Non-uniform lighting produces a spatial brightness gradient across the image that has nothing to do with the actual objects present, so a single global correction cannot fix all regions correctly.

(b) Key issues:

- Illumination varies smoothly or abruptly across the frame
- Global thresholding/enhancement fails because the 'correct' brightness differs per region
- Foreground/background separation becomes unreliable in unevenly lit areas

(c) Applied technique — Adaptive Histogram Equalization (CLAHE) / Homomorphic filtering / Adaptive (local) thresholding: Divide the image into local tiles (or work in the log-frequency domain) and equalize/threshold each region using its own local statistics rather than one global value.

(d) Justification: Local/adaptive methods correct contrast and brightness per neighbourhood, adapting to the illumination gradient, whereas a single global correction (e.g. plain histogram equalization) cannot compensate for spatially varying lighting.

(e) Summary: Cause → non-uniform lighting; Fix → local/adaptive correction; Result → normalized illumination across the whole image.

3. Salt-and-Pepper Noise

(a) Interpretation: Salt-and-pepper noise is impulsive: a random subset of pixels is forced to extreme values (0 or 255) while the rest of the document remains correct, corrupting text readability with scattered black/white speckles.

(b) Key issues:

- High-magnitude, isolated random pixel spikes
- Averaging-type filters would blend noise into neighbouring text and blur characters
- Noise density can vary, requiring a filter robust to outliers

(c) Applied technique — Median Filtering: Replace each pixel by the median (middle ranked value) of its 3×3 (or larger) neighbourhood.

(d) Justification: The median filter is preferred over mean/averaging filters because impulsive noise values sit at the extremes of the sorted neighbourhood and are excluded from the middle rank, so they are completely rejected rather than blended in — this removes the noise without blurring text edges.

(e) Summary: Noise type → impulsive; Fix → median filtering; Result → clean, readable document with edges/text preserved.

4. Blurred Edges

(a) Interpretation: Smoothing filters attenuate rapid intensity changes to suppress noise, but genuine edges are also rapid intensity changes — so an over-sized or non-edge-aware smoothing kernel removes real structural detail along with the noise.

(b) Key issues:

- Kernel size too large relative to feature scale
- Isotropic (non-edge-aware) filtering treats edges and noise the same way
- Loss of structural detail needed for downstream analysis

(c) Applied technique — Unsharp Masking / Laplacian sharpening (or switch to an edge-preserving filter such as the bilateral filter): Add back a scaled high-frequency (edge) component to the smoothed image, or re-filter the original using an edge-aware method that smooths flat regions while respecting strong gradients.

(d) Justification: Unsharp masking directly restores the high-frequency detail attenuated by smoothing; alternatively, bilateral filtering avoids the trade-off altogether by weighting neighbours on both spatial distance and intensity similarity, so it smooths within regions but not across edges.

(e) Summary: Cause → indiscriminate smoothing; Fix → sharpening / edge-aware filtering; Result → noise suppressed while edges remain sharp.

5. High-Frequency Noise

(a) Interpretation: Fine-grained noise corresponds to high spatial-frequency random variations superimposed on the image, commonly introduced by atmospheric interference or sensor electronics, and it can visually mask small genuine features.

(b) Key issues:

- Noise frequency band overlaps with the frequency band of real fine features
- Naive spatial filtering risks removing genuine detail along with noise
- Requires precise, frequency-selective removal

(c) Applied technique — Frequency-domain Low-Pass Filtering (Gaussian or Butterworth LPF via the Fourier Transform): Transform the image to the frequency domain, attenuate components above a chosen cutoff frequency, then inverse-transform back to the spatial domain.

(d) Justification: A Gaussian or Butterworth low-pass filter is preferred over an ideal (brick-wall) filter because its smooth transition avoids ringing artefacts, giving controlled noise suppression while retaining as much genuine low/mid-frequency structure as possible.

(e) Summary: Cause → high-frequency noise; Fix → frequency-domain LPF; Result → cleaner image with genuine features still visible.

6. Object Boundary Detection

(a) Interpretation: Accurately locating the boundary between different tissue types is essential for diagnosis, measurement, and treatment/surgical planning; the boundary corresponds to an intensity transition between adjacent tissue classes.

(b) Key issues:

- Low contrast between certain tissue types
- Noise in acquisition (e.g. MRI/CT artefacts)
- Blurred or gradual transitions rather than sharp edges
- Overlapping intensity ranges of different tissues

(c) Applied technique — Canny Edge Detection: Apply Gaussian smoothing, compute intensity gradients, perform non-maximum suppression, then use hysteresis (dual) thresholding to trace connected edges.

(d) Justification: Canny is well suited here because its built-in smoothing step reduces noise sensitivity, non-maximum suppression produces thin, well-localized edges, and hysteresis thresholding links weak-but-genuine edges to strong ones — giving accurate, continuous boundaries critical for medical use.

(e) Summary: Requirement → accurate tissue boundaries; Fix → Canny edge detection; Result → precise, noise-robust boundary map for diagnosis.

7. Simple Segmentation

(a) Interpretation: Since object and background intensities are well separated, this is a straightforward segmentation case with a clearly bimodal intensity distribution.

(b) Key issues:

- Histogram shows two well-separated peaks (bimodal)
- No significant overlap between object and background intensity ranges

(c) Applied technique — Global Thresholding using Otsu's Method: Automatically compute the threshold that minimizes intra-class variance (equivalently maximizes inter-class variance) between the two intensity groups, then binarize the image at that value.

(d) Justification: Because the histogram is clearly bimodal, a single global threshold is sufficient, and Otsu's method removes the need for manual/subjective threshold selection by choosing it statistically from the histogram itself.

(e) Summary: Case → simple, well-separated intensities; Fix → Otsu global thresholding; Result → clean binary segmentation of object vs background.

8. Complex Segmentation

(a) Interpretation: When several regions' intensity ranges overlap, no single threshold value can correctly separate all of them, making simple thresholding unreliable.

(b) Key issues:

- Overlapping histograms across multiple regions
- No single boundary value cleanly separates all classes
- Simple intensity-based rules misclassify overlapping pixels

- (c) Applied technique — K-means Clustering / Watershed Segmentation / Graph-based methods (e.g. GrabCut):** Group pixels using multiple features (intensity plus spatial/gradient information) rather than a single global cut-off, or treat the gradient image as a topographic surface (watershed) to separate regions at ridge lines.
- (d) Justification:** Unlike a single threshold, clustering and watershed-based methods use additional information (spatial proximity, local gradients, or feature vectors) to disambiguate pixels whose raw intensities overlap, correctly handling the complexity that simple thresholding cannot.
- (e) Summary:** Cause → overlapping intensities across regions; Fix → clustering/watershed segmentation; Result → correctly separated regions despite intensity overlap.

9. Broken Edges

- (a) Interpretation:** Edge maps often contain gaps where the gradient magnitude briefly falls below the detection threshold (due to noise or a locally weak intensity change), even though the underlying boundary is continuous.
- (b) Key issues:**
- Strict/high threshold cutting off weak-but-genuine edge segments
 - Noise causing local gradient dips along a true boundary
 - Non-maximum suppression occasionally removing borderline pixels
- (c) Applied technique — Edge Linking via Hysteresis Thresholding (Canny) or Morphological Closing:** Use two thresholds so weak edge pixels connected to strong ones are retained (hysteresis), and/or apply morphological dilation followed by erosion (closing) to bridge small gaps.
- (d) Justification:** Hysteresis thresholding recovers genuinely-connected weak edges that a single strict threshold would discard, and morphological closing directly bridges small remaining gaps — together restoring continuity without introducing excessive noise-based edges.
- (e) Summary:** Cause → gaps from thresholding/noise; Fix → hysteresis + morphological closing; Result → continuous, closed object boundaries.

10. Combined Requirement

- (a) Interpretation:** The system has two competing goals at once: suppress sensor/lighting noise, while keeping the fine edges that indicate defects fully intact — a plain smoothing-then-sharpening approach risks reintroducing noise or losing detail.
- (b) Key issues:**
- Standard smoothing blurs the very edges needed for defect detection
 - Standard sharpening amplifies any remaining noise
 - Order of operations affects final quality
- (c) Applied technique — Edge-preserving denoising (Bilateral filter or Median filter) followed by controlled sharpening (Unsharp Masking):** First denoise using a filter that respects edges, then apply a mild sharpening step to reinforce defect boundaries.
- (d) Justification:** Bilateral/median filtering removes noise while explicitly preserving strong intensity discontinuities (edges); performing denoising before sharpening also ensures the later sharpening step does not amplify noise that a naive sharpen-first approach would.
- (e) Summary:** Requirement → noise removal with edge preservation; Fix → denoise-then-sharpen sequence; Result → clean image with defect edges intact for reliable detection.

11. Low Brightness Image

- (a) Interpretation:** In low light, most pixel intensities are clustered near zero, so subtle differences in dark regions are indistinguishable and overall visibility is poor.
- (b) Key issues:**

- Histogram skewed heavily toward the dark end
- Loss of differentiation between shadow-region details
- Underexposed sensor response

(c) Applied technique — Gamma Correction ($\gamma < 1$) or Log Transformation / Histogram Equalization: Apply $s = c \cdot r^\gamma$ with $\gamma < 1$ (or $s = c \cdot \log(1+r)$) to expand the dark intensity range proportionally more than the bright range.

(d) Justification: Gamma correction with $\gamma < 1$ (or a log transform) boosts low-intensity pixels non-linearly by a larger relative amount than high-intensity pixels, lifting overall visibility without saturating the few brighter pixels present.

(e) Summary: Cause → underexposure; Fix → gamma/log brightening; Result → visible shadow detail and improved overall brightness.

12. Overexposed Image

(a) Interpretation: Overexposure pushes many pixel values close to or at the maximum (255), clipping highlight detail so that different bright surfaces become indistinguishable.

(b) Key issues:

- Saturated/clipped highlight region
- Reduced effective dynamic range among bright pixels
- Loss of texture/detail in bright areas

(c) Applied technique — Gamma Correction ($\gamma > 1$) or bright-range contrast compression: Apply $s = c \cdot r^\gamma$ with $\gamma > 1$ to compress the bright end of the range, spreading out the previously clipped/near-saturated values.

(d) Justification: Gamma correction with $\gamma > 1$ compresses high intensities more strongly than low ones, pulling clustered near-white values apart so that lost highlight detail becomes visible again, correcting the overexposure.

(e) Summary: Cause → saturation/clipping; Fix → gamma darkening ($\gamma > 1$); Result → recovered highlight detail and better balanced exposure.

13. Gaussian Noise

(a) Interpretation: Gaussian noise is additive random noise following a normal distribution, applied independently to every pixel, and it is most visible as graininess in otherwise smooth/flat regions.

(b) Key issues:

- Noise spread continuously across the whole image (not isolated pixels)
- Reduced signal-to-noise ratio, obscuring subtle features
- Requires statistical averaging rather than outlier rejection

(c) Applied technique — Gaussian Smoothing Filter (or linear averaging filter): Convolve the image with a Gaussian-weighted kernel so nearby pixels are averaged, reducing the random per-pixel variance introduced by the noise.

(d) Justification: Because Gaussian noise is additive and continuously distributed (not impulsive), a linear averaging/Gaussian filter is statistically effective at reducing its variance; a median filter, by contrast, is optimized for impulsive outliers and is less efficient against this noise type.

(e) Summary: Noise type → additive Gaussian; Fix → Gaussian smoothing; Result → reduced graininess with detail statistically preserved.

14. Speckle Noise

(a) Interpretation: Speckle is a multiplicative, signal-dependent noise typical of coherent imaging systems (ultrasound, SAR), appearing as a granular texture pattern whose intensity scales with the underlying signal, rather than being simply added on top.

(b) Key issues:

- Multiplicative (not additive) noise model, so generic additive-noise filters are suboptimal
- Needs to preserve diagnostically-important edges/boundaries while filtering
- Noise intensity varies with the signal itself

(c) Applied technique — Speckle-specific filters — Lee filter, Frost filter, or anisotropic diffusion: Use a filter that adapts its smoothing strength to local signal variance/statistics, modelling the multiplicative noise explicitly rather than assuming additive noise.

(d) Justification: Lee/Frost-type filters are specifically derived from the multiplicative speckle noise model and adjust filtering strength per-region based on local variance, giving better speckle suppression with edge preservation than generic linear filters designed for additive noise.

(e) Summary: Noise type → multiplicative speckle; Fix → Lee/Frost or anisotropic diffusion filtering; Result → reduced granularity with diagnostic edges preserved.

15. Loss of Fine Details

(a) Interpretation: Fine detail loss indicates that a prior filtering step (smoothing/denoising) was too aggressive, removing genuine high-frequency content along with noise.

(b) Key issues:

- Oversized smoothing kernel relative to the scale of fine features
- Non-edge-aware filtering applied uniformly everywhere
- Cumulative blur from multiple processing stages

(c) Applied technique — Unsharp Masking / High-Boost Filtering: Subtract a blurred version of the image from the original to isolate high-frequency content, then add this back (scaled) to the processed image to restore detail.

(d) Justification: High-boost filtering explicitly reintroduces a scaled high-frequency component that was attenuated by earlier smoothing, recovering perceptible detail; using smaller kernels or edge-aware filters in the original processing chain would prevent the loss from recurring.

(e) Summary: Cause → over-smoothing earlier in the pipeline; Fix → high-boost/unsharp restoration; Result → recovered fine detail without reintroducing the original noise.

16. Edge Enhancement Requirement

(a) Interpretation: Downstream feature extraction needs prominent, well-defined edges as reliable landmarks; faint or low-contrast edges in the raw image would degrade extraction accuracy.

(b) Key issues:

- Weak/faint edges due to low contrast or noise
- Insufficient emphasis of high-frequency structure in the raw image

(c) Applied technique — High-pass filtering / Laplacian sharpening / Sobel gradient magnitude: Convolve the image with a Laplacian or gradient (Sobel) mask to amplify local intensity discontinuities corresponding to edges.

(d) Justification: Laplacian/high-pass masks respond strongly and specifically to rapid intensity change (edges) while producing near-zero response in flat regions, making genuine edges stand out clearly for whatever feature-extraction algorithm follows.

(e) Summary: Requirement → strong, extractable edges; Fix → Laplacian/Sobel-based enhancement; Result → clearly defined edges ready for feature extraction.

17. Background Noise Removal

(a) Interpretation: Random variations in the background can be misclassified as foreground (or vice versa) during segmentation, reducing accuracy by introducing spurious regions.

(b) Key issues:

- Noise pixels occasionally crossing the segmentation threshold
- Fragmented/noisy segmentation mask
- False positive small regions

(c) Applied technique — Pre-segmentation smoothing (median/Gaussian filter) plus post-segmentation

Morphological Opening: Smooth the image before thresholding to reduce random variance, then apply erosion followed by dilation on the binary mask to remove small isolated noise blobs.

(d) Justification: Smoothing before segmentation lowers the chance that noise alone crosses the threshold, while morphological opening afterward specifically removes any small spurious regions that remain, together improving segmentation accuracy from both directions.

(e) Summary: Cause → background noise crossing threshold; Fix → pre-smoothing + morphological opening; Result → cleaner segmentation mask with noise-induced regions removed.

18. Threshold Selection Problem

(a) Interpretation: Manual threshold selection is unreliable when the histogram is not clearly bimodal or when illumination varies, since no single fixed value can be confidently chosen by inspection.

(b) Key issues:

- Ambiguous or shallow valley between histogram peaks
- Illumination variation shifting the ideal threshold across the image
- Subjective, non-repeatable manual selection

(c) Applied technique — Otsu's Method (automatic global threshold) or Adaptive/Local Thresholding:

Compute the threshold statistically as the value minimizing intra-class variance (Otsu), or compute a local threshold per neighbourhood if illumination varies.

(d) Justification: Otsu's method removes subjectivity by deriving the threshold directly and optimally from the image histogram; if a single global value still proves inadequate due to uneven lighting, adaptive/local thresholding further improves robustness.

(e) Summary: Problem → unclear/inconsistent threshold; Fix → Otsu / adaptive thresholding; Result → objective, reliable segmentation threshold.

19. Texture-Based Segmentation

(a) Interpretation: When regions share similar average intensity but differ in spatial pattern (texture), intensity-based methods have no basis to separate them, since they only look at pixel value, not spatial arrangement.

(b) Key issues:

- Overlapping/similar mean intensity across regions
- Thresholding-based methods are blind to spatial pattern differences
- Need descriptors that capture local structure, not just value

(c) Applied technique — Texture-based segmentation using GLCM features, Local Binary Patterns (LBP), or Gabor filter banks combined with clustering/classification: Extract texture descriptors per neighbourhood (contrast, homogeneity, LBP codes, or Gabor responses) and cluster/classify pixels or blocks based on these descriptors.

(d) Justification: Texture descriptors encode spatial relationships between neighbouring pixels rather than a single intensity value, allowing regions with identical average intensity but different patterns to be correctly separated — something purely intensity-based thresholding cannot achieve.

(e) Summary: Case → texture-differentiated regions; Fix → texture feature extraction + clustering; Result → correct segmentation based on pattern, not just intensity.

20. Multiple Object Detection

(a) Interpretation: When several distinct objects share similar intensity to each other (even if separable from the background), simple intensity thresholding groups them into a single blob rather than distinguishing individual objects.

(b) Key issues:

- Touching or overlapping objects with matching intensity
- No intensity-based cue exists to separate individual instances
- Need shape/spatial information in addition to intensity

(c) Applied technique — Watershed Segmentation (with distance transform) or Connected Component

Analysis after morphological separation: Treat the (inverted) distance-transformed binary mask as a topographic surface and 'flood' from local maxima/minima so touching objects are split at their natural boundary ridges.

(d) Justification: Watershed segmentation uses topological/gradient structure rather than raw intensity alone, so it can separate touching objects at the ridge between them, succeeding precisely where simple thresholding merges them into one region.

(e) Summary: Case → multiple similar-intensity objects; Fix → watershed/distance-transform segmentation; Result → individually separated object instances.

21. Edge Noise Problem

(a) Interpretation: Gradient-based edge detectors are inherently sensitive to noise, since differentiation amplifies small random intensity fluctuations, producing spurious edge pixels that do not correspond to real object boundaries.

(b) Key issues:

- High sensitivity of gradient operators to pixel-level noise
- False/spurious edges cluttering the true boundary map
- Difficulty distinguishing real weak edges from noise-induced ones

(c) Applied technique — Pre-smoothing (Gaussian filter) before edge detection, and/or Canny's built-in smoothing with hysteresis thresholding: Smooth the image to suppress small random fluctuations before computing gradients, then use dual (hysteresis) thresholds to retain only genuinely connected edges.

(d) Justification: Smoothing before differentiation reduces the noise-driven fluctuations that gradient operators would otherwise amplify, and hysteresis thresholding further discriminates weak noise-induced responses from weak-but-genuine edges connected to strong ones.

(e) Summary: Cause → noise amplified by differentiation; Fix → pre-smoothing + hysteresis thresholding; Result → cleaner edge map with noise-induced false edges suppressed.

22. Region Merging Requirement

(a) Interpretation: Over-segmentation has produced multiple small regions that actually belong to the same underlying object, so they need to be combined into a single, correctly-sized region.

(b) Key issues:

- Excessive initial splitting from a sensitive segmentation algorithm
- Minor intensity variance treated as a separate region
- No consideration of inter-region similarity/adjacency

(c) Applied technique — Region Merging using a Region Adjacency Graph (RAG): Build a graph of adjacent regions, compare their mean intensity (or other descriptors), and merge adjacent region pairs whose difference falls below a chosen tolerance.

(d) Justification: Comparing similarity between adjacent regions (rather than judging each region in isolation) ensures merges reflect genuine structural relationships, correctly correcting the fragmentation left by the initial over-sensitive segmentation.

(e) Summary: Cause → fragmented over-segmented regions; Fix → adjacency-based region merging; Result → coherent regions that correctly represent each true object.

23. Over-Segmentation

(a) Interpretation: Over-segmentation occurs when the algorithm splits the image into far more regions than there are true meaningful objects, typically because it is overly sensitive to small intensity/texture variations or noise.

(b) Key issues:

- Oversensitive splitting criteria or too many watershed markers/seeds
- Noise treated as legitimate region boundaries
- Increased computational and interpretive overhead from excess regions

(c) Applied technique — Region Merging post-processing, or relaxed segmentation parameters (larger growing/merging threshold, fewer seed markers), optionally preceded by smoothing: Merge adjacent statistically similar regions after segmentation, and/or increase the algorithm's tolerance threshold or pre-smooth to reduce spurious initial splits.

(d) Justification: Loosening similarity thresholds or merging statistically similar neighbours directly counteracts oversensitivity, reducing the excess region count while still respecting genuine boundaries.

(e) Summary: Cause → algorithm oversensitivity/noise; Fix → merging / parameter relaxation; Result → a sensible number of regions matching the true objects present.

24. Under-Segmentation

(a) Interpretation: Under-segmentation occurs when distinct, meaningful regions are incorrectly combined into one because the segmentation criterion was too coarse or the tolerance too loose to detect the subtle difference between them.

(b) Key issues:

- Similarity/tolerance threshold set too high
- Weak boundaries between genuinely different regions
- Segmentation not fine-grained enough to capture subtle intensity/texture differences

(c) Applied technique — Finer/multi-level thresholding, reduced region-growing tolerance, or edge-guided region splitting: Tighten the similarity criterion (lower allowed intensity difference) or use additional boundary/edge information to force a split where it was wrongly omitted.

(d) Justification: Tightening the similarity threshold makes the algorithm respect subtler intensity distinctions that were previously ignored, correctly separating regions that had been wrongly merged together.

(e) Summary: Cause → overly loose merging criterion; Fix → tighter thresholds / edge-guided splitting; Result → correctly separated, meaningful regions.

25. Boundary Refinement

(a) Interpretation: Boundaries produced directly from pixel-level edge detection can be jagged or imprecise due to noise, pixel discretization, or a coarse detection method, reducing the accuracy of the extracted shape.

(b) Key issues:

- Pixel-level staircase/jagged artefacts
- Noise-induced local irregularities along the boundary
- Lack of sub-pixel precision in the raw edge output

(c) Applied technique — Active Contours (Snakes), Morphological smoothing, or polygon approximation (e.g. Douglas–Peucker): Iteratively adjust boundary points to minimize an energy function balancing smoothness with fidelity to the actual image edges, converging on a refined contour.

(d) Justification: Active contour models explicitly trade off smoothness against accuracy through their energy formulation, producing a refined, smooth boundary that better represents the true object shape than the raw, noise-affected pixel-level output.

(e) Summary: Cause → rough/imprecise raw boundary; Fix → active contour refinement; Result → smooth, accurate boundary representation.

26. Frequency-Based Enhancement

(a) Interpretation: The desired enhancement (e.g. removing periodic noise or applying a global frequency-selective adjustment) is more naturally and efficiently expressed and applied in the frequency domain than as an equivalent spatial-domain operation.

(b) Key issues:

- Spatial filters may require very large kernels to achieve an equivalent global effect
- Periodic noise patterns are hard to isolate spatially but appear as distinct peaks in the frequency spectrum
- Computational cost of large spatial kernels

(c) Applied technique — Fourier Transform based filtering — Low/High-pass or Notch filtering: Transform the image with the FFT, directly manipulate (attenuate/boost) the relevant frequency components (or remove specific periodic-noise peaks via a notch filter), then inverse transform.

(d) Justification: Operating directly in the frequency domain allows exactly the frequencies responsible for the unwanted pattern (or the ones to be preserved/boosted) to be targeted precisely and efficiently, which is difficult to replicate with an equivalently-sized spatial kernel.

(e) Summary: Requirement → global/periodic frequency-selective correction; Fix → FFT-based filtering; Result → efficient, targeted enhancement not easily achieved spatially.

27. Mixed Noise Scenario

(a) Interpretation: Two different noise types are present simultaneously — continuous additive Gaussian noise and impulsive salt-and-pepper noise — so no single filter optimized for just one type will fully clean the image.

(b) Key issues:

- Median filtering alone does not efficiently reduce continuously-distributed Gaussian variance
- Averaging/Gaussian filtering alone does not remove impulsive outliers (and is skewed by them)
- Requires a combined or sequential strategy

(c) Applied technique — Sequential filtering — Median filter first, then Gaussian smoothing (or an adaptive hybrid filter): Apply a median filter to remove impulsive outliers first, then apply Gaussian smoothing to reduce the remaining continuously distributed noise.

(d) Justification: Removing impulsive outliers first (median filter, which is unaffected by them) prevents them from skewing the subsequent smoothing step; the Gaussian smoothing that follows then appropriately addresses the remaining additive noise — addressing both noise types in the order best suited to each.

(e) Summary: Noise type → mixed Gaussian + impulsive; Fix → median then Gaussian filtering; Result → both categories of noise substantially reduced.

28. Edge Preservation Filtering

(a) Interpretation: Standard smoothing filters inherently trade off noise removal against edge sharpness, since both noise and edges correspond to local intensity variation that linear filters treat identically.

(b) Key issues:

- Linear filters blur edges along with noise
- Insufficient smoothing leaves noise; excessive smoothing destroys edges
- Need a filter that distinguishes 'edge' from 'noise' variation

(c) Applied technique — Bilateral Filtering or Anisotropic (Perona–Malik) Diffusion: Weight neighbouring pixels by both spatial closeness and intensity similarity (bilateral), or diffuse intensity preferentially within homogeneous regions while inhibiting diffusion across strong gradients (anisotropic diffusion).

(d) Justification: Because these filters explicitly incorporate intensity similarity (not just spatial proximity) into their weighting, they average within homogeneous regions to suppress noise but avoid averaging across strong intensity discontinuities, directly satisfying both requirements at once.

(e) Summary: Requirement → noise removal with edge preservation; Fix → bilateral filter / anisotropic diffusion; Result → smoothed regions with edges left intact.

29. Region Growing Failure

(a) Interpretation: Region growing's outcome depends heavily on where the seed is placed; if the seed lands on noise, a boundary pixel, or an otherwise unrepresentative location, the growth criterion propagates from a bad starting statistic and produces an incorrect region.

(b) Key issues:

- High sensitivity of the algorithm to the initial seed
- A poorly-placed seed's error propagates through the entire growth process
- No built-in validation of seed suitability

(c) Applied technique — Improved seed selection — choose seeds from verified homogeneous interior areas (low local variance), use multiple seeds with voting/averaging, or automatic seed selection via histogram-peak analysis: Validate candidate seed locations against local variance before growing, or grow from several seeds and combine/cross-check the resulting regions.

(d) Justification: Selecting seeds from low-variance, clearly representative interior pixels (or combining multiple seeds) reduces reliance on any single potentially poor choice, making the region-growing process considerably more robust and reliable.

(e) Summary: Cause → poorly chosen seed; Fix → validated/multiple seed selection; Result → correct, reliable region growth outcome.

30. Industrial Inspection Enhancement

(a) Interpretation: The system must reliably reveal small defects despite sensor/lighting noise, requiring a complete pipeline rather than any single processing step, since enhancement, denoising, and segmentation each address a different part of the problem.

(b) Key issues:

- Noise obscuring subtle defect signatures
- Low contrast between defects and surrounding material
- Need for a fast, robust, repeatable automated pipeline

(c) Applied technique — Full pipeline: (1) noise reduction (median/bilateral filter) → (2) contrast enhancement (histogram equalization / CLAHE) → (3) edge or threshold-based detection (Canny / Otsu) → (4) morphological refinement to isolate defect regions: Apply each stage in sequence so every step operates on progressively cleaner, more distinguishable data.

(d) Justification: Sequencing denoising, then enhancement, then detection, then refinement ensures each stage benefits from the improvements of the previous one, maximizing overall defect-detection accuracy and reliability in an automated industrial setting — a single isolated technique could not address all the requirements at once.

(e) Summary: Requirement → robust automated defect detection in noisy images; Fix → denoise → enhance → segment → refine pipeline; Result → reliable, structured defect detection suitable for industrial inspection.

Code of Conduct:

I G Kushwanth (192424150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.kushwanth